

Vol 11 Chapter 5 — D_IV⁵ Geometry: Coset, Boundary, Coordinates

Keeper (author pass — deep math/physics revision)

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Chapter 5 — D_IV⁵ Geometry: Coset, Boundary, Coordinates

Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Level 1 — one sentence

The substrate domain D_{IV}^5 is realized explicitly as a bounded subset of $\mathbb{C}^5$ (Lie ball), with topological boundary having interior structure (Shilov boundary, characteristic submanifolds) — and the electron in BST is identified as a “Shilov-boundary primitive cycle” (Casey’s Saturday work) tying particle physics to substrate geometry.

Level 2 — graduate-physicist precision

5.1 Lie ball realization

D_{IV}^5 in Lie ball realization: $z = (z_1, \dots, z_5) \in \mathbb{C}^5$ satisfies

$$1 - 2z^*z + |z^T z|^2 > 0, \quad |z^T z| < 1$$

Open bounded subset of $\mathbb{C}^5$. Real dim 10.

5.2 Coset structure

$$D_{IV}^5 = SO_0(5, 2)/[SO(5) \times SO(2)].$$

$SO_0(5, 2)$ acts transitively on D_{IV}^5 by biholomorphisms (combined with cocycle).

Base point $z = 0$: stabilizer = $SO(5) \times SO(2)$.

5.3 Topological boundary

Topological boundary $\partial D_{IV}^5 = \overline{D_{IV}^5} \setminus D_{IV}^5$ has 9-real-dimensional interior. Not all of it is “boundary” in the function-theoretic sense.

5.4 Shilov boundary

The Shilov boundary $S \subset \partial D_{IV}^5$ is the smallest closed set on which holomorphic functions attain maximum modulus.

For D_{IV}^5 : Shilov boundary is a 5-real-dimensional submanifold, the unique closed orbit of $SO_0(5, 2)$ acting on ∂D_{IV}^5 .

The Shilov boundary $S \cong U(2) \cdot S^4$ or similar (depending on convention) — explicit identification depends on coordinate choice.

5.5 Casey’s Saturday: electron as Shilov primitive cycle

Memory: Casey’s Saturday work identifies the electron as a substrate “Shilov-boundary primitive cycle” (Task #53 completed).

Substrate-mechanism: the electron K-type configuration is the substrate’s natural primitive (= minimal non-trivial) topological cycle on the Shilov boundary of D_{IV}^5 .

Pin(2) half-integer-weight K-type structure (Vol 5 Ch 3) restricted to Shilov boundary gives the electron’s spin-1/2.

Charge = $SO(2)$ weight (Task #56 + T2470).

5.6 Other particle BST identifications

Casey’s Saturday work also identified: - Proton = bulk geodesic minimum-energy K-type (Task #54) - Neutrino = $SO(2)$ -trivial winding (Task #55) - Electric charge = $SO(2)$ weight (Task #56)

Plus the “Three Commitments Principle” as Saturday’s framework (Task #57).

5.7 Characteristic subboundary

Beyond Shilov: D_{IV}^5 has Bergman-Šilov characteristic submanifolds organized by $SO_0(5, 2)$ orbit structure. Each orbit corresponds to a distinct substrate K-type “extremal” configuration.

5.8 K-audit anchors

- Task #53 completed: Electron = Shilov-boundary primitive cycle
- Task #54, #55, #56, #57: Saturday particle identifications

- Task #71: SP-26 W-31 asymmetric ontology — baryons primary

Level 3 — 5th-grader accessibility

D_{IV}^5 as bounded subset of $\mathbb{C}^5$: looks like a “Lie ball” — explicit inequality $1 - 2z^*z + |z^T z|^2 > 0$. Shilov boundary: special part of the boundary where holomorphic functions take their max. For D_{IV}^5 : a 5D submanifold of the 9D topological boundary. Casey’s Saturday work: the electron is the substrate’s “primitive cycle on the Shilov boundary” — the smallest non-trivial topological loop on this boundary. Proton = bulk geodesic minimum-energy K-type. Neutrino = $SO(2)$ -trivial winding. Electric charge = $SO(2)$ weight. The substrate geometry IS particle physics.

What comes next

Chapter 6 develops K3 surfaces, Niemeier lattices, and the Leech lattice.

Where to look this up

- Faraut and Koranyi 1990
- BST Tasks #53-#57; T2470